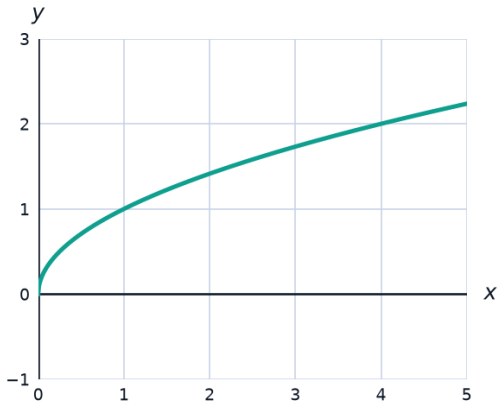
Integration

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

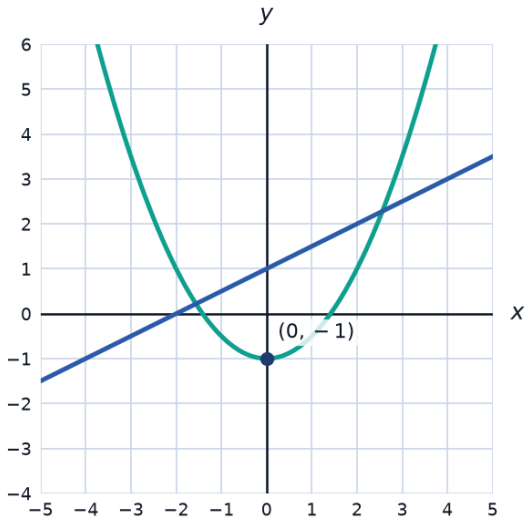
1 [Foundational] • Target: Volumes of revolution: the disk method

The region under $y = \sqrt{x}$ on $[0, 4]$ is revolved about the x-axis. Find the volume generated, using $V = \pi \int y^2 \, dx$.



2 [Foundational] • Target: Integration by substitution

Find $\int 2x(x^2 + 1)^4 \, dx$ using the substitution $u = x^2 + 1$.



3 [Foundational] • Target: Integration by substitution

Find $\int x\sqrt{x^2 + 9} \, dx$ using the substitution $u = x^2 + 9$.

4 [Foundational] • Target: Integration by substitution

Find $\int (2x + 1)(x^2 + x)^3 \, dx$ using the substitution $u = x^2 + x$.

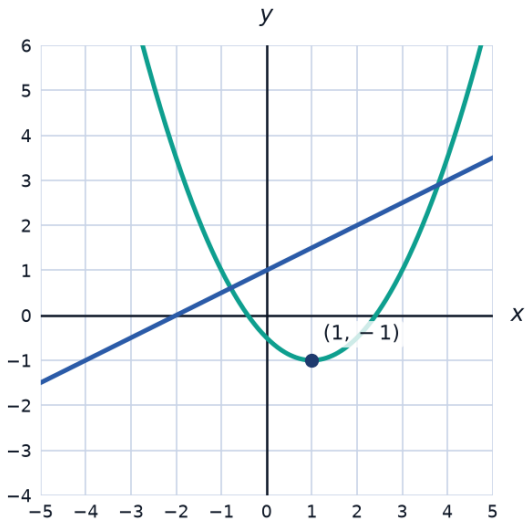
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5 [Foundational] • Target: Integration by substitution

Find $\int 3x^2(x^3 - 2)^4\,dx$ using the substitution $u = x^3 - 2$.

6 [Foundational] • Target: Integration by parts

Find $\int xe^x\,dx$ using integration by parts.



7 [Foundational] • Target: Integration by parts

Find $\int x\cos x\,dx$ using integration by parts.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration by parts

Find $\int \ln x\,dx$ using integration by parts with $u = \ln x$ and $dv = dx$.

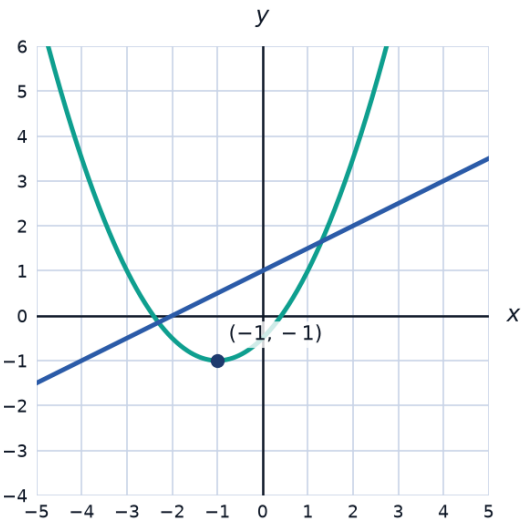
9 [Intermediate] • Target: Integration by parts

Find $\int x^2e^x\,dx$ using integration by parts twice.

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10 [Intermediate] • Target: Integration using partial fractions

Find $\int \frac{5x+7}{(x+1)(x+2)}\,dx$, first decomposing into partial fractions.



11 [Intermediate] • Target: Integration using partial fractions

Find $\int \frac{1}{x^2-4}\,dx$ using partial fractions, factoring $x^2-4=(x-2)(x+2)$.

12 [Intermediate] • Target: Integration using partial fractions

Find $\int \frac{3x-1}{x(x-3)}\,dx$, first decomposing into partial fractions.

13 [Intermediate] • Target: Integration using partial fractions

Find $\int \frac{2x+3}{(x-1)(x+4)}\,dx$, first decomposing into partial fractions.

14 [Intermediate] • Target: Integrating trigonometric, exponential and logarithmic functions

Find $\int \sin(3x)\,dx$ and $\int e^{2x}\,dx$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

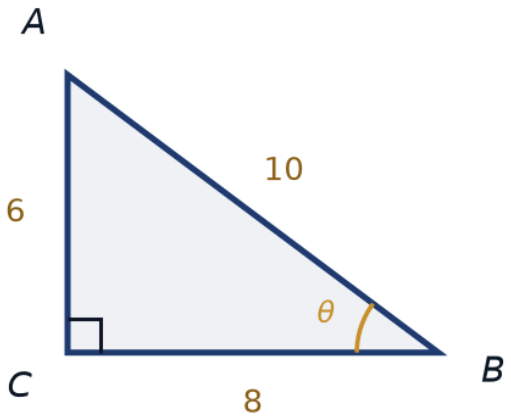
15 [Challenging] • Target: Integrating trigonometric, exponential and logarithmic functions • Cross-Lesson: 6.01 Sequences & Series

Find $\int \frac{4x}{x^2+1}\,dx$.

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16 [Challenging] • Target: Integrating trigonometric, exponential and logarithmic functions • Cross-Lesson: 7.01 Vectors

Find $\int \tan x \, dx$, using the substitution $u = \cos x$.



17 [Challenging] • Target: Integrating trigonometric, exponential and logarithmic functions • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

Find $\int \frac{1}{2x+3} \, dx$.

18 [Challenging] • Target: Definite integrals using substitution • Cross-Lesson: 9.01 Complex Numbers

Evaluate $\int_0^2 x(x^2+1)^3 \, dx$ using the substitution $u = x^2 + 1$.

19 [Challenging] • Target: Definite integrals using substitution • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Evaluate $\int_0^{\pi/2} \sin x \cos^3 x \, dx$ using the substitution $u = \cos x$.

20 [Challenging] • Target: Definite integrals using substitution • Cross-Lesson: 10.01 Numerical Methods

Evaluate $\int_1^e \frac{\ln x}{x} \, dx$ using the substitution $u = \ln x$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Standard Integrals (Trig, Exp & Rational)

Find $\int \frac{2}{3x-1} \, dx$.

a. $\frac{2}{3}\ln|3x-1| + C$

b. $2\ln|3x-1| + C$

c. $\frac{1}{3}\ln|3x-1| + C$

d. $-\frac{2}{(3x-1)^2} + C$

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22 [Foundational] • Target: Standard Integrals (Trig, Exp & Rational)

Find $\int \cos^2(x)\,dx$.

a. $\frac{x}{2} + \frac{\sin(2x)}{4} + C$

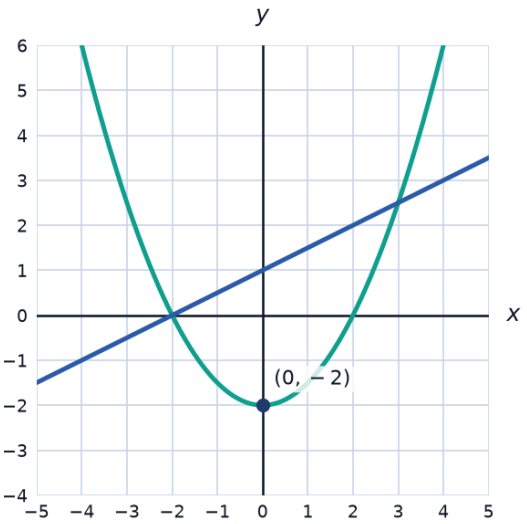
b. $\frac{\sin^3(x)}{3} + C$

c. $\frac{x}{2} - \frac{\sin(2x)}{4} + C$

d. $\sin(2x) + C$

23 [Foundational] • Target: Standard Integrals (Trig, Exp & Rational)

Evaluate $\int_0^{\pi/4} \tan(x)\,dx$.



a. $\frac{1}{2}\ln 2$

b. $\ln 2$

c. 1

d. $\frac{\sqrt{2}}{2}$

24 [Foundational] • Target: Integration by Substitution

Evaluate $\int x\sqrt{x^2+5}\,dx$.

a. $\frac{1}{3}(x^2+5)^{3/2} + C$

b. $\frac{1}{2}(x^2+5)^{3/2} + C$

c. $\frac{2}{3}(x^2+5)^{3/2} + C$

d. $(x^2+5)^{3/2} + C$

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25 [Foundational] • Target: Integration by Substitution

Evaluate $\int_0^1 \frac{x}{x^2+1}\,dx$.

a. $\frac{1}{2}\ln 2$

b. $\ln 2$

c. $\frac{\pi}{4}$

d. 1

26 [Foundational] • Target: Integration by Substitution

Use the substitution $x = 2\sin\theta$ to find $\int \sqrt{4-x^2}\,dx$.

a. $2\theta + \sin(2\theta) + C$

b. $4\theta + 2\sin(2\theta) + C$

c. $2\arcsin(x/2) + C$

d. $\frac{1}{2}(4-x^2)^{3/2} + C$

27 [Foundational] • Target: Integration by Parts

Find $\int xe^{3x}\,dx$.

a. $\frac{xe^{3x}}{3} - \frac{e^{3x}}{9} + C$

b. $\frac{xe^{3x}}{3} - \frac{e^{3x}}{3} + C$

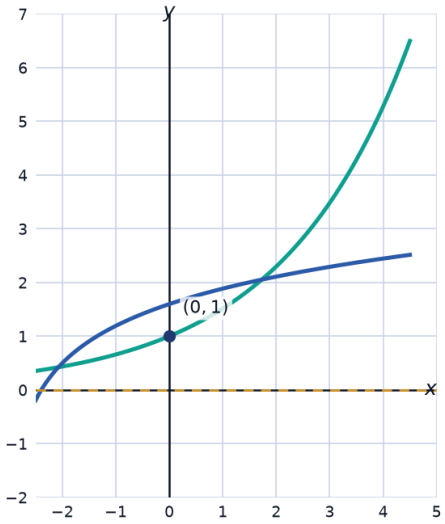
c. $xe^{3x} - e^{3x} + C$

d. $\frac{x^2e^{3x}}{6} + C$

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28 [Foundational] • Target: Integration by Parts

Evaluate $\int_1^e x\ln x\,dx$.



a. $\frac{e^2+1}{4}$

b. $\frac{e^2-1}{4}$

c. $\frac{e^2}{2}$

d. $\frac{e^2-1}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Integration by Parts

Evaluate $\int_0^\pi x\sin x\,dx$.

a. π

b. 2π

c. 0

d. 2

30 [Intermediate] • Target: Partial Fractions Integration & Trapezium Rule

Find $\int \frac{5x-2}{(x-2)(x+1)}\,dx$.

a. $\frac{8}{3}\ln|x-2|+\frac{7}{3}\ln|x+1|+C$

b. $3\ln|x-2|+2\ln|x+1|+C$

c. $8\ln|x-2|+7\ln|x+1|+C$

d. $\ln|(x-2)(x+1)|+C$

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31 [Intermediate] • Target: Partial Fractions Integration & Trapezium Rule

Use the trapezium rule with 4 intervals (5 ordinates) to approximate $\int_1^3 \frac{1}{x}\,dx$.

a. 1.117

b. 1.099

c. 1.150

d. 1.050

32 [Intermediate] • Target: Partial Fractions Integration & Trapezium Rule

Is the trapezium rule approximation in the previous question an overestimate or underestimate, and why?

a. Overestimate, because the curve $y = 1/x$ is concave up

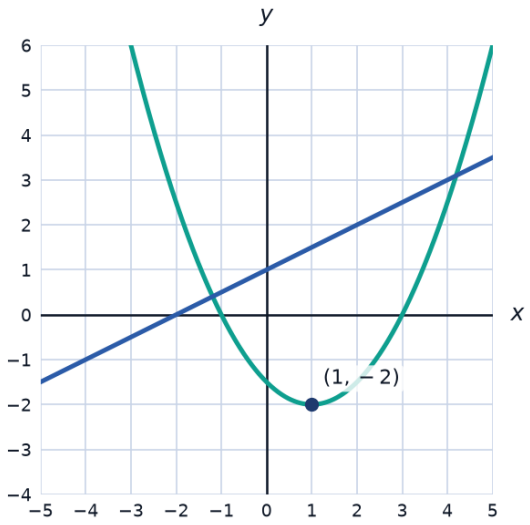
b. Underestimate, because $y = 1/x$ is decreasing

c. Overestimate, because the intervals are too wide

d. Underestimate, because the chords lie below the curve

33 [Intermediate] • Target: Integration by substitution

Using the trapezium rule with 5 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 15, y_1 = 13, y_2 = 3, y_3 = 15, y_4 = 10, y_5 = 6$.



a. $\frac{619}{12}$

b. $\frac{105}{2}$

c. 31

d. $\frac{103}{2}$

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34 [Intermediate] • Target: Integration by parts

Using the trapezium rule with 4 strips of width $h = 3$, estimate $\int y\,dx$ given $y_0 = 8, y_1 = 18, y_2 = 5, y_3 = 18, y_4 = 17$.

a. $\frac{1927}{12}$

b. $\frac{321}{2}$

c. 99

d. $\frac{327}{2}$

35 [Intermediate] • Target: Integration using partial fractions

Using the trapezium rule with 6 strips of width $h = 3$, estimate $\int y\,dx$ given $y_0 = 4, y_1 = 3, y_2 = 19, y_3 = 3, y_4 = 16, y_5 = 2, y_6 = 17$.

a. $\frac{1927}{12}$

b. $\frac{321}{2}$

c. 96

d. $\frac{327}{2}$

36 [Intermediate] • Target: Integrating trigonometric, exponential and logarithmic functions

Using the trapezium rule with 6 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 9, y_1 = 1, y_2 = 12, y_3 = 20, y_4 = 11, y_5 = 17, y_6 = 12$.

a. $\frac{145}{2}$

b. $\frac{859}{12}$

c. 41

d. $\frac{143}{2}$

37 [Intermediate] • Target: Definite integrals using substitution

Using the trapezium rule with 4 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 16, y_1 = 15, y_2 = 14, y_3 = 19, y_4 = 7$.

a. $\frac{715}{12}$

b. $\frac{71}{2}$

c. $\frac{121}{2}$

d. $\frac{119}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Volumes of revolution: the disk method • Cross-Lesson: 9.01 Complex Numbers

Using the trapezium rule with 3 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 1, y_1 = 1, y_2 = 2, y_3 = 10$.

a. $\frac{17}{2}$

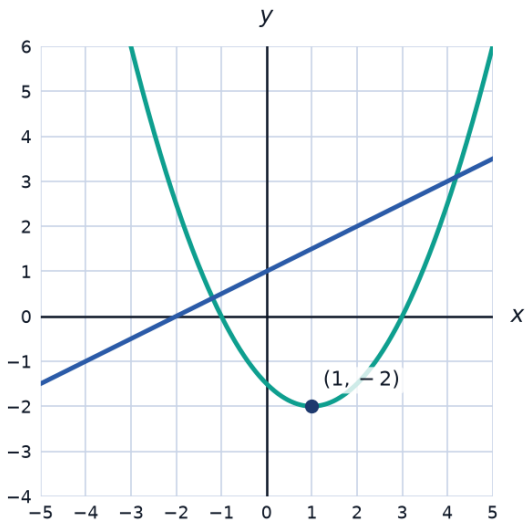
b. 7

c. $\frac{103}{12}$

d. $\frac{19}{2}$

39 [Challenging] • Target: Volumes of revolution: washer and shell methods • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Using the trapezium rule with 2 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 9, y_1 = 10, y_2 = 4$.



a. $\frac{35}{2}$

b. $\frac{23}{2}$

c. $\frac{199}{12}$

d. $\frac{33}{2}$

40 [Challenging] • Target: A well-designed word problem: designing a vase • Cross-Lesson: 10.01 Numerical Methods

Using the trapezium rule with 2 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 5, y_1 = 7, y_2 = 8$.

a. $\frac{27}{2}$

b. $\frac{163}{12}$

c. 10

d. $\frac{29}{2}$

41 [Challenging] • Target: Standard Integrals (Trig, Exp & Rational) • Cross-Lesson: 11.01 Differential Equations

Using the trapezium rule with 2 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 10, y_1 = 5, y_2 = 5$.

a. 10

b. $\frac{151}{12}$

c. $\frac{27}{2}$

d. $\frac{25}{2}$

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42 [Challenging] • Target: Integration by Substitution • Cross-Lesson: 12.01 Poisson Distribution

Using the trapezium rule with 5 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 14, y_1 = 8, y_2 = 10, y_3 = 9, y_4 = 7, y_5 = 5$.

a. $\frac{89}{2}$

b. $\frac{53}{2}$

c. $\frac{87}{2}$

d. $\frac{523}{12}$

43 [Challenging] • Target: Integration by Parts • Cross-Lesson: 13.01 Continuous Random Variables

Using the trapezium rule with 4 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 2, y_1 = 8, y_2 = 5, y_3 = 13, y_4 = 5$.

a. $\frac{33}{2}$

b. $\frac{355}{12}$

c. $\frac{61}{2}$

d. $\frac{59}{2}$

44 [Challenging] • Target: Partial Fractions Integration & Trapezium Rule • Cross-Lesson: 14.01 Sampling & Estimation

Using the trapezium rule with 4 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 7, y_1 = 11, y_2 = 6, y_3 = 13, y_4 = 12$.

a. $\frac{81}{2}$

b. $\frac{475}{12}$

c. $\frac{79}{2}$

d. $\frac{49}{2}$

45 [Challenging] • Target: Integration by substitution • Cross-Lesson: 15.01 Hypothesis Testing

Using the trapezium rule with 4 strips of width $h = 1$, estimate $\int y\,dx$ given $y_0 = 13, y_1 = 15, y_2 = 9, y_3 = 13, y_4 = 12$.

a. $\frac{99}{2}$

b. 31

c. $\frac{595}{12}$

d. $\frac{101}{2}$

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